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ORIGINAL RESEARCH

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ORIGINAL RESEARCH

New azolo benzoxazepinyl oxazolidinone substituent activity investigations

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Abstract :

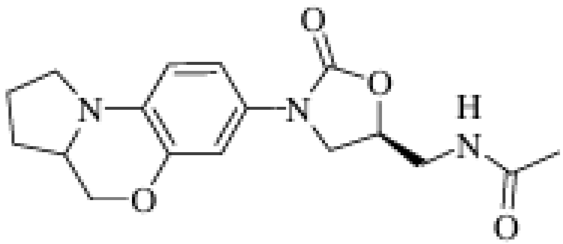
We developed a series of new tricyclic compounds that mimic the conformationally limited structure of Linezolid to identify potent antibacterial agents using the entropically favored 'bioactive conformation' strategy. Starting from tricyclic compound 1, the benzazepine derivative 2 was synthesized with increased C–N bond flexibility. However, compound 2 showed reduced antibacterial activity compared to the parent compound. In contrast, compound 3 exhibited promising antibacterial activity, which may be attributed to the rigidity provided by the pyrrole ring. The synthesis of these compounds, structure–activity relationship (SAR) studies and the biological evaluation of lead compound 3 are presented.

Keywords: Tricyclic compounds, Linezolid, Antibacterial agents, Bioactive conformation, Benzazepine derivative, C–N bond flexibility, Pyrrole ring, Antibacterial activity, Structure Activity Relationship (SAR), Lead compound, Drug design, Compound synthesis

Introduction:

Current antibiotic therapy are threatened by Gram-positive bacteria's drug resistance. MRSA, VRE, and GISA are now more than just scientific curiosity, posing a life-threatening threat to physicians worldwide. "Super bugs" are here to stay, and in addition to many drug resistance control measures, a concentrated effort is needed to create new antibiotics with unique mechanisms of action. Oxazolidino--nes, a new class of synthetic antibacterials, Inhibit bacterial protein biosynthesis at the 50S ribosomal subunit and are active against resistant Gram-positive organisms like MRSA VRE, and GISA. an early stage. Pharmacia introduced the first medicine in this family, Linezolid, in 2000. Despite its antibacterial promise and unique method of action, no oxazolidinone molecule, except Linezolid, reached the market. Additionally, antibiotic resistance is inevitable, and Linezolid is no exception. Linezolid is not suitable for longterm therapy due to myelosuppression, however some patients have experienced no major side effects after taking it for over 2 years. Therefore, powerful and safer oxazolidinone molecules must be developed. In our prior comm--unication, we highlighted how bioactive conformation in a medicinal molecule helps it bind to a target favorably, delivering entropic benefit. Every molecule and medication should rotate freely along bonds. A drug has several energy states (conformations), but only the bioactive form binds to the receptor. To attach to the receptor, a flexible drug molecule must limit its rotation along bonds and freeze to the bioactive state, and which is entropically unfavorable. Constraining the medication molecule restricts rotation along bonds and if bioactive compounds are found, powerful medicine can be given. To support this idea, we created a structurally constrained tricyclic molecule (1,4) with limited rotation around the C-N link (Fig. 1). To expand on our bioactive conformation notion, we created Ben-zazepine derivative 2, and which featured a more flexible tricyclic structure around the C-N link than 1. However, with additional rotational freedom, the molecule was inactive. In contrast, molecule 3, an intermediate of 2, had promising antibacterial activity due to its stiff pyrrole ring. We conducted SAR on the right side of the molecule to increase activity. We present the synthesis of molecules 2 and 3, SAR studies on 3, and appraisal of 3 as a lead chemical.

Figure1 :



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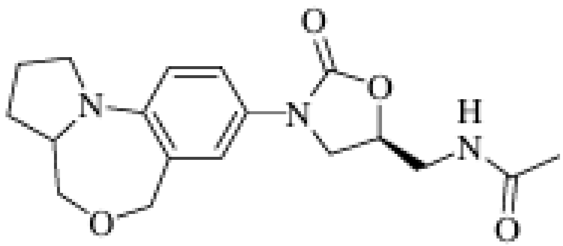

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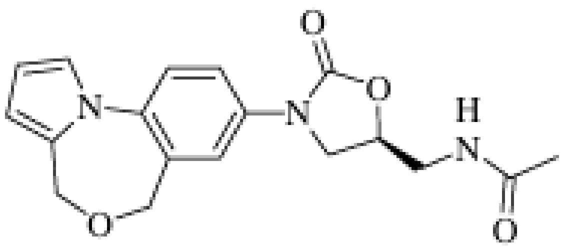

3

Synthesis of azide 13:

Nitration of isatoic anhydride 4 yields nitro 5 (Scheme 1). Methanolysis of anhydride 5 yielded amino-ester 6. By rea--cting compound 6 with 2, 5-dimethoxy tetrahydrofu ran in acetic acid, amino functionality was changed to pyrrole ring (7). Vilsmeyer Hack formylation of chemical 7 produced aldehyde 8, which was reduced with sodium borohydride and acidified to benz-azulene molecule 9. Compound 11 was formed via nitro to amine and CBzCl protection of amine, which was treated with (R)-glycidyl butyrate in BuLi to yield alcohol 12. Mesylation and sodium azide displacement transformed 12's alcohol functionality to azide 13.

Synthesis of the derivatives:

The amine 14 was reduced by azide 13 in triphenylphos--phine and water (Scheme 2) and transformed to derivatives using either matching acid chlorides in triethyl-amine or DCC and DMAP. Thioacetic acid treatment of azide 13 created acetamide 3, and which hydrogenation yielded the pyrrolidine derivative 2. Thioacetamide 27 was made by treating amine 14 with ethyldithioacetate and TEA. Alkyl--chloroformates and TEA were used to create carbamate derivatives from amine 14. The isothiocyanate 15 was produced from amine 14 using carbon disulfide, TEA, and ethylchloroformate, followed by thiocarbamate derivatives following alcohol reflux (Table 1). All compounds produced above were tested for Gram-positive antibacterial activity in vitro. The antibacterial activity of compound 2 decreased due to its greater C–N bond flexibility. However, compound 3 seems to work like Linezolid after gaining stiffness from the pyrrole ring instead of pyrrolidine. Thus, SAR experiments on amine 14 were conducted to increase antibacterial activity (Table 2). Acetamide 3 molecules with longer chains or bulkier structures had lower activity (3 versus 16–20). However, chloro acetamide compound 21 remained active. It's interesting that 22 was less active. Carbamates 24–26 made bromo derivative 22 less active. The carbamates 24–26 were less active than the acetamides. Thioamide 27 and thiocarbamate 28 were more active than acetamide 3. The analogous thiocarbamate derivative 29 was less active than 28. After SAR studies, compounds with in vitro activity comparable to Linezolid (3, 19, 21, 27, and 28) were tested in mice for pharmacokinetics and in vivo action. Table 3 shows the mean plasma concentration and ED50 of these drugs and Linezolid. Oral absorption from the mouse gastrointestinal tract was quick, and all drugs except compound 27 achieved full plasma concentration (Tmax) at 0.5 h. The data showed compound 3 outperformed others with a mean plasma concentration (Cmax) of 11.7 lg/mL and an AUC(0–1) of 21.09 lg h/mL.

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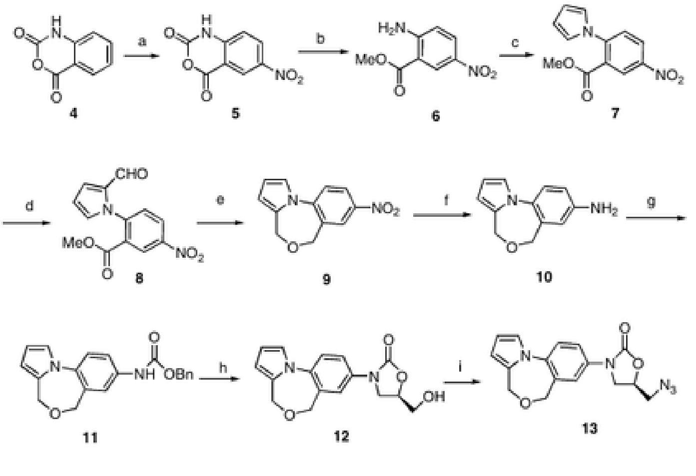
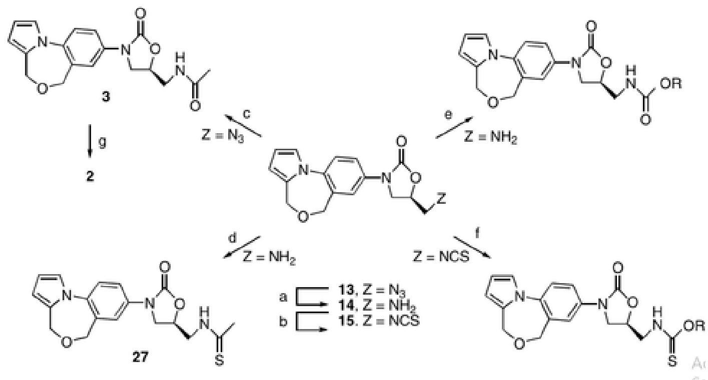
All other drugs had poor pharmacokinetics, as shown in a mouse pharmacodynamic research. In the mouse systemic infection model, compound 3 had an ED50 of 10.10 mg/kg for subcutaneous (sc) treatment and 12.5 mg/kg for per os (po) administration, compared to 2.18 mg/kg (sc) and 5.38 mg/kg (po) for Linezolid. All other drugs with ED50 > 30 mg/kg dosage failed to protect animals, which matched their poor pharmacokinetics. Wistar rats were used to study chemical 3's oral pharmacokinetics. Figure 2 and Table 4 show the pharmacokinetic characteristics and mean plasma concentration versus time profiles of chemical 3 and Linezolid. After oral administration, compound 3 achieved Tmax at 1.5 h, while Linezolid was rapidly absorbed and reached Tmax at 0.5 h, the elimination rate (Kel) and terminal half-life (t1/2) matched Linezolid. However, the Cmax of 12.9 lg/mL and AUC(0–1) of 52.90 lg h/mL were lower than Linezolid, which had a Cmax of 27.09 lg/mL and an AUC(0–1) of 83.45 lg h/mL. In conclusion, acetamide derivative 3 was the lead molecule from azolo benzoxaze--pinyl oxazolidinone SAR investigations. Thioamide and thiocarbamate replaced acetamide of 3 better in vitro. However, small alterations at position 3, such as replacing acetamide with a longer or thicker group, had negative effects on in vitro activity. The thiocompounds 27 and 28 had strong in vitro potency but poor in vivo efficacy, presumably due to weak Pharmacokinetic profiles. The chemical 3 demonstrated good in vivo effectiveness, complementing its pharmacokinetic profile. It is also noteworthy that rats had a longer systemic exposure and elimination half-life of 3 than mice.



Scheme 1. *Reagents and conditions: (a) KNO3, H2SO4, 0–5 C, 86%; (b) MeOH, NaOH, 90%; (c) 2,5-dimethoxyte--trahydrofuran, CH3COOH, 85%; (d) DMF, POCl3, 86%; (e) NaBH4, MeOH, HCl, 60%; (f) ammonium formate, 10% Pd–C, THF–MeOH, 66%; (g) CBzCl, NaHCO3, 96%; (h) n-BuLi, (R)-glycidyl butyrate, 85%; (i) MsCl/TEA and NaN3/ DMF, 92%.*



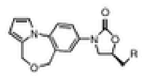
Scheme 2. *Reagents: (a) Ph3P/H2O, 80%; (b) CS2, TEA, ethylchloroformate; (c) CH3COSH; (d) ethyldithioacetate, TEA; (e)ROCOCl, TEA; (f) ROH; (g) Pd on charcoal, H2.*

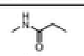
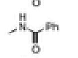
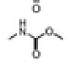
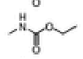
Experimental:

General:

Capillary melting point equipment recorded uncorrected melting points. 1H and 13C NMR were done at 200 and 50 MHz. The internal standard is tetramethylsilane (TMS), and chemical shifts are recorded in delta units. HP-5989A mass spectrometers recorded spectra. Combustion analysis was conducted with a Perkin-Elmer 2400 analyzer. All analytical work was done at Discovery Research's Analytical Research Department.

Table 1. SAR studieson the amine 14



Compound	R	Conditions	Yield (%)
16		Propionyl chloride TEA, DCM, 0 °C, 0.5 h	34
17		Isovaleryl chloride TEA, DCM, 0 °C, 0.5 h	41
18		Cyclopropyl carboxylic acid DCC, DMAP, $CHCl_3$, 0 °C, rt, 1 h	41
19		Acryloyl chloride TEA, DCM, 0 °C, 0.5 h	30
20		Benzoyl chloride TEA, DCM, 0 °C, 0.5 h	66
21		2-Chloroacetyl chloride TEA, DCM, 0 °C, 0.5 h	75
22		2-Bromoacetyl chloride TEA, DCM, 0 °C, 0.5 h	43
23		3,3,3-Trifluoropropionic acid DCC, DMAP, $CHCl_3$, 0 °C, rt, 1 h	49
24		Methylchloroformate TEA, DCM, 0 °C, 0.5 h	23
25		Ethylchloroformate TEA, DCM, 0 °C, 0.5 h	29
26		Benzylchloroformate TEA, DCM, 0 °C, 0.5 h	14
27		Ethyldithioacetate TEA, THF, rt, 6 h	21
28		CS_2, NEt_3, THF methylchloroformate, methanol, reflux	40
29		CS_2, NEt_3, THF methylchloroformate, ethanol, reflux	38

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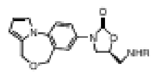
6-Nitro-1H-benzo[d][1,3]oxazine-2,4-dione :

At 0 C, potassium nitrate (12.40 g, 122.7 mmol) was added in tiny batches to a solution of isatoic anhydride 4 (20 g, 122.7 mmol) in con. H2SO4 (30 mL) over 30 min and stirred for 15 min. Filtered precipitate was generated from the reaction fluid poured onto ice flakes. The pale brown nitro compound 5 (24 g, 94%) was prepared by washing and drying the solid under vacuum. IR (KBr): 3199, 1779, 1758, 1354, 1336, 1034 cm 1. 1H NMR (DMSO-d6): d 12.31 (s, 1H), 8.53 (m, 2H), 7.31 (d, J = 10.0 Hz, 1H). MS: 209 (M++1), 191, 182, 164, 134, 106, 90.

Methyl 2-amino-5-nitro-benzoate:

6-nitro-1H-benzo[d][1,3]oxazine-2,4-dione (5,25 g, 120.2 mmol) and NaOH (480 mg, 12.0 mmol) were refluxed in methanol under N2 for around 15 minutes. The reaction mixture was cooled to rt, diluted with ethylacetate (750 mL), washed with water, dried over anhydrous Na2SO4, and concentrated to yield 6 as a yellow solid (21 g, 89%). IR (KBr): 3473, 3361, 1707, 1631, 1319 cm 1. 1H NMR (CDCl3): 8.62 (s, 1H), 8.14 (d, J = 10.0 Hz, 1H), 7.83 (br s, 1H), 6.89 (br s, 2H), 3.85, 3H. MS: 196 (M+), 180, 164, 134, 119, 106, 91.

Table 2. In vitro potency of azolo benzoxazepinyl oxazolidinones



Compound[a]	R	SA 33591	SA 49951	SA 29213	EF 29212	EF 12201	EFs 12202
1		4	8	16	16	16	16
2		32	32	32	>32	>32	>32
3		2	2	2	2	2	2
16		4	4	4	8	8	8
17		4	8	8	8	8	8
18		16	16	32	32	32	32
19		4	2	2	4	4	4
20		>32	>32	>32	>32	>32	>32
21		1	2	2	2	2	2
22		4	4	4	4	4	4
23		8	8	8	8	8	8
24		4	4	4	8	8	8
25		16	16	16	16	16	16
26		>32	>32	>32	>32	>32	>32
27		0.5	1	1	1	0.5	1
28		0.5	0.5	0.5	2	2	2
29		4	4	4	8	8	8
Linezolid		2	2	2	2	2	4

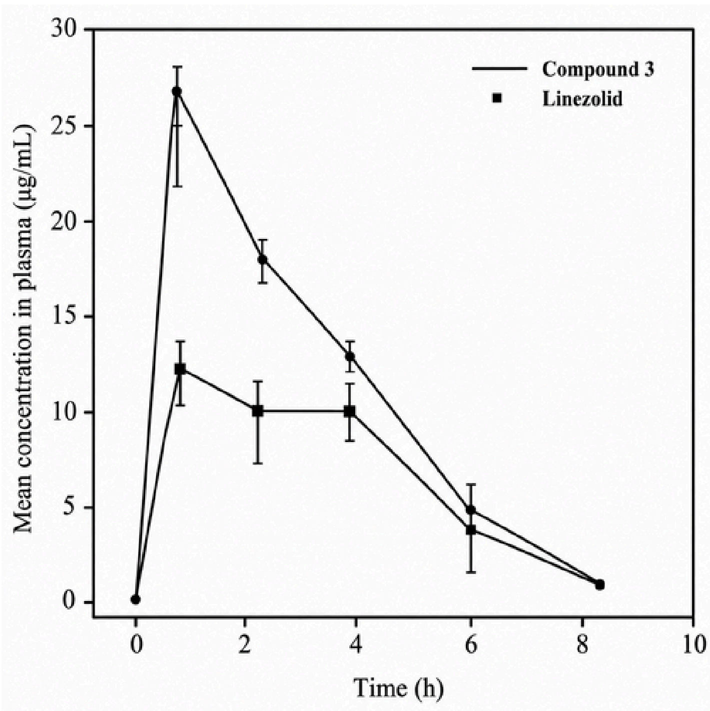
Table 3. Pharmacokinetic and pharmacodynamic parameters of selected compounds in Swiss albino mice:

Parameters	3	19	21	27	28	Linezolid
$AUC_{0-\infty}$ (μg·h/mL)	21.09	3.83	0.55	1.04	6.72	71.03
T_{max} (h)	0.5	0.5	0.5	1.0	0.5	0.5
C_{max} (μg/mL)	11.7	2.18	0.47	0.18	3.2	28.13
$t_{1/2}$ (h)	0.8	0.6	0.6	0.9	0.8	1.0
ED_{50} (mg/kg)						
sc	10.10	>30	>30	>30	>30	2.18
po	12.50	>30	>30	>30	>30	5.38

Table 4. Singledose oral pharmacokinetic parame- -ters of compound 3 and Linezolid in Wistar rats:

Parameters	Compound 3	Linezolid
Data derived from (n)	6	4
$AUC_{(0-\infty)}$ (μg h/mL)	52.90 ± 19.53	84.66 ± 12.17
C_{max} (μg/mL)	12.9 ± 4.07	27.09 ± 2.12
T_{max} (h)	1.50 ± 0.58	0.50 ± 0.00
K_{el} (h^{-1})	0.58 ± 0.21	0.61 ± 0.06
$t_{1/2}$ (h)	1.31 ± 0.40	1.15 ± 0.12

Figure 2. Single dose oral pharmacokinetic parameters of compound 3 and Linezolid in Wistar rats.



Methyl 5-nitro-2-pyrrol-1-yl-benzoate :

To a solution of methyl 2-amino-5-nitro-benzoate 6 (21 g, 107.1 mmol) in acetic acid (200 mL), add a solution of 2,5-dimethoxy tetrahydrofuran (14.16 g, 107.1 mmol) in acetic acid (500 mL). Heat the combination at reflux under N2 for 2 hours.

Cooling the reaction mixture and pouring it into water yielded compound 7 as a viscous liquid (28 g, 85%). IR (Neat): 1741, 1502, 1335, 738 cm 1. 1H NMR (CDCl3): 8.60, 8.41, 7.45, 6.85, 6.18, 3.80 (s, 3H). MS: 246 (M+), 188.

Methyl 2- (2- formyl- pyrrol-1-yl) -5- nitro -benzoate :

Under N2 at 0 C, 17 g of phosphorus oxychloride (110.9 mmol) was added dropwise to 250 mL of DMF to create a viscous yellow complex. To the reaction mixture, a solution of methyl 5-nitro-2-pyrrol-1-yl-benzoate (26 g, 105.7 mmol) was added in DMF at the same temperature for 0.5 hours. The reaction mixture was stirred for 2 hours after reaching rt. It was then poured into Na2CO3-based water. After filtering, washing, and drying under vacuum, the precipitate yielded a white solid (21.6 g, 80%). IR (KBr): 1720, 1661, 1344, 767 cm 1. 1H NMR (CDCl3): d = 9.45 (s, 1H), 8.85 (s, 1H), 8.40 (d, J = 7.0 Hz, 1H), 7.35 (d), 7.11 (s), 6.95 (s), 6.45 (m), 3.75 (s, 3H). MS: 274 (M+), 246, 215, 169, 140.

8- Nitro- 4H, 6H-5- oxa- 10b- aza- benzo [e] azulene :

Sodium borohydride (690 mg, 18.3 mmol) was slowly added to a solution of methyl 2-(2-formyl-pyrrol-1-yl)-5-nitro-benzoate 8 (5.00 g, 18.3 mmol) in MeOH (40 mL). After stirring the reaction mixture for 1 h, 10 mL con. HCl was added and diluted with 100 mL water. The reaction mixture was extracted with 100 mL 3• ethylacetate after 10 min of stirring. The organic layer was washed with water, brine, dried over anhydrous Na2SO4, and concentrated to yield nitro compound 9 as a yellow solid (2.50 g, 60%). KBr IR: 1H NMR (CDCl3): d 8.18 (d, J = 10.0 Hz, 2H), 7.58 (d, J = 5.0 Hz, 1H), 7.11 (s, 1H), 6.40 (s, 2H), 4.58 (d, J = 10.0 Hz, 4H). MS: 230 (M+), 201, 183, 154, 127.

4H, 6H-5-Oxa-10b -aza-benzo [e] azulen-8-ylamine :

Add 10% Pd–C (500 mg) to an ice-cold solution of 8-nitro-4H,6H-5-oxa-10b-aza-benzo[e]azulene 9 (1.00 g, 4.4 mmol) and ammo-nium formate (1.09 g, 17.4 mmol) in THF–MeOH (1:4, 15 mL) and stir at rt under N2 for 1 h. Geltrate was concentrated by filtering the reaction mixture over a Celite pad. The residue was diluted with 100 mL ethylacetate and washed with water and brine. After drying over anhydrous Na2SO4, it was concentrated to yield amine 10 as a pale brown solid (580 mg, 66%). IR (KBr): 3356, 2855, 1668, 1625, 1512, 1310, 1268, 1064, 720 cm 1. 1H NMR (CDCl3): d 7.20 (d, J = 1.6 Hz, 1H), 7.01 (s, 1H), 6.64–6.68 (m, 3H), 6.24 (s, 2H), 4.45 (s, 2H), 4.25 (s, 2H). MS: 200 (M+), 171, 154, 144, 130, 115, 106, 91.

Benzyl (4H, 6H- 5 -oxa- 10b- aza- benzo [e] azulen-8-yl)- carbamate :

A solution of amine 10 (4 g, 20.0 mmol) in acetone (20 mL) was mixed with a solution of NaHCO3 (1.69 g, 40.0 mmol) in water (10 mL). To the reaction mixture, add 50% benzylchloroformate (3.75 g, 22.0 mmol) dropwise and cool to 0 C for 0.5 hours under N2. After reaching rt, the reaction mixture was put into 30 mL water. The precipitate was filtered and dried under vacuum to yield the compound 11 as an o-white solid (6.00 g, 90%). IR (KBr): 3250, 1728, 1565, 1513, 1317, 1236 cm 1. 1H NMR (CDCl3): d 7.28 (m, 8H), 7.01 (s, 1H), 6.78 (br s, 1H), 6.25 (s. 2H), 5.20 (s, 2H), 4.42 (d, J = 10.0 Hz, 4H). MS: 334 (M+), 199, 169, 91.

(R)-5-Hydroxymethyl-3-(4H,6H-5-oxa-10b -aza-benzo[e]azulen-8-yl)-oxazolidin-2 one :

Benzyl (4H,6H-5-oxa-10b-aza-benzo [e]azulen-8-yl)-carba-mate 11 (8.00 g, 24.0 mmol) was dissolved in dry THF (100 mL) and n-BuLi (15% solution in hexane, 15.33 mL, 36.0 mmol) was added at 78 C over 0.5 h and stirred for another 0.5 h. After quenching with saturated NH4Cl solution, the reaction mixture was diluted with 300 mL of ethylacetate and water. The organic layer was separated and the aqueous layer was extracted using 300 mL of ethylacetate (2•). The organic layer was brine-washed and dried over anhydrous Na2SO4. Alcohol 12 was crystallized from ethylacetate-petroleum ether residual, resulting in an o-white solid (6.10 g, 85%). IR (KBr): 3398, 1726, 1511, 1062 cm 1. 1H NMR (CDCl3): 7.71-7.74 (m, 1H), 7.59-7.43 (d, J = 8.9 Hz, 1H), 7.04-7.43 (s, 1H), 6.32 (6H), 4.75-4.80 (m, 1H), 4.48 (d, J = 9.4 Hz, 4H), 4.01-4. MS: 301 (M++1), 257, 176.

(R)-5-Azidomethyl-3-(4H,6H-5-oxa-10b-aza - benzo[e]azulen-8-yl)-oxazolidin-2-one :

To a cooled (0 C) alcohol 12 solution in CH2Cl2 (100 mL), add triethyl-amine (4.11 g, 41.0 mmol) and slowly add methanesulfonylchloride (3.49 g, 30.5 mmol) over 30 min. The organic layer was washed with water, brine, and dried (Na2SO4) after dilution with CH2Cl2 (100 mL). Concentration yielded an off-white solid mesylate (7.07 g). Add sodium azide (3.61 g, 55.6 mmol) to a solution of mesylate (7.00 g, 18.5 mmol) in DMF (50 mL) and heat to 80 C for 1 hour. Next, the reaction mixture was extracted with ethylacetate (200 mL 3•), rinsed with water (200 mL 2•), brine, and dried with anhydrous Na2SO4.

A 5.7 g, 95% solid of the chemical was produced by evaporating volatiles. KBr = IR CDCl3 1H NMR: d 7.71-7.74 (m, 1H), 7.57 (s, 1H), 7.43 (d, J = 8.6 Hz, 1H), 7.01 (s, 1H), 6.32 (s, 2H), 4.79-4.85 (m, 1H), 4.47 (d, J = 9.4 Hz, 4H), 4.14 (t, J = 8.9 Hz, 1H), 3.90-3.94 (m, 1H), 3.59-3.76 (m, 2H). MS: 326 (M++1), 300, 298, 201.

(S)-N- [3-(4H,6H-5-Oxa-10b -aza-benzo [e] azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-acetamide (3) :

Under nitrogen atmosphere, (R)-5-Azidomethyl-3-(4H,6H-5-oxa-10b-aza-benzo[e]azu-len-8-yl)-oxazolidin-2-one 13 (1.2 g, 3.7 mmol) was treated with 8 mL of thiolacetic acid. The product 3 was purified on silica gel (100-200 mesh) using a methanol-chloroform (1:9) mixture, yielding a white solution (700 mg, 58%). IR (KBr): 3298, 1737, 1679, 1512, 1321, 1059, 736 cm. 1H NMR (CDCl3): d 7.72–7.43 (m, 3H), 7.06 (m, 3H). 13C NMR (DMSO-d6): 170.0, 154.1, 136.6, 135.5, 129.7, 129.5, 121.5, 120.9, 120.0, 119.5, 109.3, 109.2, 71.6, 66.1, 59.6, 47.3, 41.4, 22.4. MS: 341 (M+), 312, 297, 254, 209, 154, 127, 105, 91.

N-[2-Oxo-3-(2,3,3a,4-tetrahydro-1H,6H-5-oxa-10b-aza-benzo [e] azulen-8-yl) -oxazoli din- 5-ylmethyl]-acet-amide (2) :

Hydrogenation of amide 3 (100 mg, 0.3 mmol) and 10% Pd–C (31 mg, 10 mol%) in ethylacetate took 16 h at 50 psi. After filtering the catalyst over Celite, the resulting solution was concentrated to obtain compound 2 (82 mg, 82%) as a colorless solid. Mp 262–264 C. IR (KBr): 3315, 1746, 1510, 752 cm 1. 1H NMR (CDCl3): d = 8.21 (t, J = 5.8 Hz, 1H), 7.38-7.31 (m, 2H), 6.89 (d, J = 8.9 Hz, 1H), 4.72-4.64 (m, 2H), 4.28 (d, J = 13.2 Hz, 1H), 4.09-4.03 (m, 1H), 3.88 (dd, J = 2.4 and 11.6 Hz, 1H), 3.72-3.67 (m, 1H), 3.39 (t, J = 5.5 Hz, 2H). 13C NMR: 169.9, 154.2, 145.4, 130.6, 130.3, 120.0, 118.7, 115.4, 75.2, 73.4, 71.3, 62.6, MS: 346 (M++1), 296. Anal. Calcd for C18H23N3O4: C: 62.59; H: 6.71; N: 12.17. Found: C: 63.11; H: 6.63; N: 12.03.

(R)-5- Aminomethyl -3- (4H,6H-5 -oxa-10b-aza-benzo [e]azulen-8-yl)-oxazolidin-2-one:

To a solution of azide 13 (7.0 g, 21.5 mmol) in THF (60 mL), triphenylphosphine (6.22 g, 23.7 mmol) was added. After 2 hours of stirring, add 1 mL of water and boil the mixture to 40-50°C for 16 hours. Water was azeotropically removed with benzene after solvent evaporation. After purifying the residue on a silica gel column (100-200 mesh) with acetone–chloroform (2:3), the sticky amine 14 (5.5 g, 77.62%) was obtained. 3386, 1748 cm IR (KBr).

1. 1H CDCl3 NMR: d = 7.72-7.75 (m, 1H), 7.58 (s, 1H), 7.42 (d, J = 8.86 Hz, 1H), 7.04, 6.32, 4.68-4.74 (m, 1H), 4.47 (d, J = 9.1 Hz, 4H), 4.09 (t, J = 8.7 Hz, 1H), 3.90-3.94 (3H), 2.98-3.17 (2H). MS: 300 (M++1), 279, 270.

(S)-N- [3- (4H,6H-5-Oxa-10b-aza-benzo [e] azulen-8-yl) -2-oxo-oxazolidin-5- ylmethyl] propiona-mide :

To a solution of amine 14 (100 mg, 0.3 mmol) in chloroform (10 mL), add triethylamine (93 lL, 0.7 mmol) at 0 C. Add propionyl chloride (70.52 mg, 0.5 mmol) and stir for 30 min at the same temperature. Dilution with chloroform and multiple washings with water and brine were done. After drying over anhydrous sodium sulfate, it was concentrated. Purifying the residue on a silica gel column (100-200 mesh) with acetone-chloroform (2:3) eluent yielded compound 16 (80 mg, 34%) as a white solid. Mp 256–258 C. IR (KBr): 3441, 3295, 2924, 1737, 1678, 1513, 732 cm 1. 1H NMR (CDCl3, 200 MHz): d 7.70–7.64 (m, 2H), 7.42 (d, J = 8.5 Hz, 1H), 7.04 (m, 1H), 6.33 (d, J = 2.0 Hz, 2H), 6.17 (br s, 1H), 4.82 (br s). 50-MHz CDCl3 13C NMR: MS: 356 (M+), 312. Anal. Calcd for C19H21N3O4: C, 64.21; H, 5.96; N, 11.82. C: 64.77; H: 5.77; N: 12.07.

(S)-N- [3- (4H,6H-5-Oxa-10b-aza-benzo [e] azulen-8-yl) -2-oxo-oxazolidin-5-ylmethyl]-isovalery lamide :

The compound was prepared using isovaleryl chloride instead of propionyl chloride, yielding 41% yield: mp 206 C. IR (KBr): 3423, 3305, 2959, 2924, 1736, 1671, 1513, 736 cm 1H NMR (CDCl3, 200 MHz): d 7.69–7.65 (m, 1H), 7.62 (d, J = 2.4 Hz, 1H), 7.45 (d, J = 13C NMR (50 MHz, CDCl3): d 173.7, 154.5, 136.7, 136.1, 130.5, 130.1, 121.5, 120.4, 119.6, 109.6, 72.1, 66.9, 60.4, 47.6, 41.7, 29.6, 26.0, 22.3. MS: 383 (M+), 339, 213, 127, 91. Calcd for C21H25N3O4: C, 65.78; H, 6.57; N, 10.96. Found: C 65.05, H 6.56, N 10.36.

(S)-[3-(4H,6H-5-oxa-10b-aza-benzo[e]azu--len-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-cyc lopropanamide

The amine 14, cyclopropane carboxylic acid (0.03 mL, 0.3 mmol), DCC (67 mg, 0.3 mmol), and DMAP (41 mg, 0.3 mmol) in chloroform were stirred for 1 h at rt. The residue was purified on silica gel (100–200 mesh) column using acetone–chloroform (2:3) as eluent to obtain the title compound 18 (50 mg, 41%). Mp 240 C. KBr: 3298, 1736, 1670, 1555, 1513, 1320, 1237, 1062 cm 1. The 1H NMR (CDCl3 + DMSO-d6) 200 MHz: d 8.37 (1H), 7.67 (2H, J =

11.2 Hz). 7.45 (d, J = 8.3 Hz, 1H), 7.09 (s), 6.26 (s, 2H), 4.79 (1H br), 4.40 (4H s), 4.13 (1H t, J = 9.0 Hz), 3.88–3.81 (m, 1H), 3.55 (d, J = 4.4 Hz, 2H), 1.61 (br s, 1H), 0.79–0.63 (m, 4H). 13C NMR (DMSO, 50 MHz): d 173.6, 154.2, 136.7, 135.6, 129.8, 129.6, 121.6, 121.0, 120.2, 120.0, 109.4, 109.3, 72.0, 66.1, 59.6, 47.4, 40.3, 13.5, 6.5. MS: 367 (M+), 323,119, 92. Anal. Calculated for N11.44; C65.38; H5.76; C20H21N3O4. Found: C, 64.79; H, 5.41; N, 11.19.

3-(4H,6H-5-Oxa-10b-aza-benzo[e]azulen-8 -yl)-2-oxo-oxazolidin-5 ylmethyl]acrylamide:

The amide 16 procedure was repeated with acryloyl chloride instead of propionyl chloride to get the title chemical 19 in 60% yield. Mp 220–221 C. KBr: 3287, 1737, 1677, 1513, 1321, 1062, 735 cm 1. 1H NMR (CDCl3 + DMSO-d6, 200 MHz): d 7.64 (t, J = 9.5 Hz, 2H), 7.43 (d, J = 8.3 Hz, 1H), 7.06 (s), 6.28 (d, J = 6.3 Hz, 3H), 5.62 (t, J = 5.9 Hz, 1H), 4.94–4.71 (m, 1H), 4.45 (d, J = 3.9 Hz, 3H), 4.14 (t, J = 8.8 Hz, 1H), 3.92 (t, J = 7.8 Hz, 1H), 3.70 (br s, 2H), 2.95. 13C NMR (50 MHz, DMSO): d 165.4, 154.0, 136.5, 135.6, 131.2, 129.7, 129.5, 125.6, 123.6, 121.3, 120.7, 120.1, 119.4, 109.2, 71.5, 66.1, 59.6, 47.4, 41.5, 40.8, 40.3. MS: 353 (M+), 309, 213, 154, 97. Anal. Calculated for C19H19N3O4: C 64.56; H 5.42; N 11.89. Found: C, 64.88; H, 5.69; N, 11.42.

S-N-[3- (4H,6H-5-Oxa-10b-aza benzo[e]azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-benzamide :

The title compound 20 was produced in 49% yield by replacing propionyl chloride with benzoyl chloride in the amide 16 procedure. Mp 234–236 C ("colorless solid"). 3336, 1738, 1663, 1540, 1515, 1319, 1233, 1054, 734 cm 1. 1H NMR (CDCl3, 200 MHz): d 7.70–7.64 (m, 2H), 7.42 (d, J = 8.5 Hz, 1H), 7.04 (m, 1H), 6.33 (d, J = 2.0 Hz, 1H). 2H, 6.17 (br s, 1H), 4.82, 4.47 (d, J = 4.4 Hz, 1H), 4.12 (t, 8.9 Hz, 1H), 3.91–3.83 (m, 1H), 3.71–3.67 (m, 2H), 2.32–2.27 (m, 2H), 1.14 (t, J = 7.6 Hz, 3H). 13C NMR (50 MHz, CDCl3): d 167.1, 154.2, 136.6, 135.5, 134.0, 131.4, 129.7, 129.5, 128.3, 127.3, 121.5, 120.9, 120.7, 119.4, 109.4, 109.3, 71.4, 66.1, 59.6, 47.6, 42.3. MS: 403 (M+), 359, 213, 147, 105, 91. Anal. Calculated for C23H21N3O4: C, 68.47; H, 5.25; N, 10.42. Found: 69.01 C, 5.05 H, 10.17 N.

S-2-Chloro-N-[3-(4H,6H-5-oxa-10b-aza-benzo[e]azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-acetamide :

The title chemical was produced in 75% yield by replacing propionyl chloride with chloroacetyl chloride in the amide 16 procedure. Mp 160–162 C (solid pale brown). 3293, 1736, 1661, 1511, 1409, 1224, 716 cm 1.

1H NMR (CDCl3) 200 MHz): 7.71 (dd, J = 2.4 and 8.8 Hz, 1H), 7.58 7.46 (d, J = 8.8 Hz, 1H), 7.08–7.06 6.35 (d, J = 2.0 Hz, 2H), 4.90–4.84 (m, 1H), 4.50 (d, J = 4.9 Hz, 4H), 4.18 (t, 9.3 Hz, 1H), 4.11 (s, 2H), 3.90–3.65 (m, 3H). 13C NMR (50 MHz, CDCl3): d 167.2, 154.3, 136.8, 135.8, 129.9, 129.7, 121.7, 121.2, 120.4, 119.8, 109.6, 109.5, 71.6, 66.3, 59.8, 47.5, 42.7, 42.0. MS: 376 (M+), 342, 298. Anal. Calculated for C18H18N3O4Cl: 57.58 C; 4.83 H; 11.20 N. Found: C, 57.87; H, 4.46; N, 10.89.

2S-Bromo-N-[3-(4H,6H-5-oxa-10b-aza-benzo[e]azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-acetamide :

A solution of amine 14 (300 mg, 1.0 mmol) in chloroform (3 mL) was treated with DCC (248 mg, DMAP (61.3 mg, 1.2 mmol) and the Stirred reaction mixture at 0 C for 30 min. Bromoacetic acid (167 mg, 1.2 mmol) was added at the same temperature and stirred for 1 h. The reaction mixture was diluted. water then chloroform (100 mL 2•). Anhydrous sodium sulfate dried the organic layer after brine washing. The solvent Evaporated residue was purified on a 100-200 mesh silica gel column with acetone-chloroform (2:3) to get compound 22 (60 mg). 43%). KBr: 3254, 1737, 1718, 1514, 890, 732 cm 1. CDCl3, 200 MHz 1H NMR: d 7.66 (dd, 7.58 (d, J = 5.9 Hz, 1H), 7.43 (d, J = 8.3 Hz, 1H), 7.04 (br s, 2H), 6.33 (s, 2H), 4.82 (m, 1H), 4.47 (d, J = 4.4 Hz, 4H), 4.14 (t, J = 9.1 Hz, 1H), 3.89–3.68 (m, 5H). 13C NMR (50 MHz, CDCl3): d 136.0, 135.6, 129.6, 129.3, 120.9, 120.0, 119.2, 109.0, 78.4, 77.8, 77.1, 71.0, 66.2, 59.6, 47.0, 41.6, 28.1. MS: 421 (M++1), 392, 341, 237, 209. 154. Anal. Calculated for C18H18BrN3O4: C, 51.44; H, 4.32; N, 10.00. C: 50.91; H: 4.44; N: 10.19.

(S)-3,3,3-Tri-fluoro-N-[3-(4H,6H-5-oxa-10b-aza-benzo[e]azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-propionic acid :

The chemical was produced in 49% yield utilizing trifluoro propionic acid instead of bromoacetic acid, as reported in 22. Mp 203–205 C. KBr: 3271, 1738, 1695, 1513, 1321, 1238, 1128, 749 cm 1. CDCl3 + DMSO-d6, 200 MHz 1H NMR: 7.66 (d, J = 8.8 Hz, 2H), 7.47 (t, J = 4.2 Hz, 1H), 7.09 (s, 1H), 6.34 (2H), 4.85 (1H), 4.49 (d, 4.18 (t, J = 9.0 Hz, 1H), 3.89 (t, J = 8.0 Hz, 1H), 3.67 (d, 4.4 Hz, 2H), 3.36 (s, 3.15 (q, J = 7.1 and 20.5 Hz, 2H). 13C NMR (50 MHz, CDCl3): d 163.6, 163.5, 154.1, 136.6, 135.6, 129.8, 129.6, 121.5, 120.9, 120.1, 119.5, 109.4, 109.3, 71.5, 66.1, 50.6, 47.2. MS: 409 (M+), 380, 213, 154, 91. Anal. Calculated for C19H18F3N3O4: C, 55.75; H, 4.43; N, 10.26; Found: C, 56.23; H, 4.36; N, 9.98.

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S-methyl-[3-(4H,6H-5-oxa-10b-aza-benzo[e]azu-len-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-carbamate :

To a solution of amine 14 (250 mg, 0.8 mmol) in DCM, triethylamine (127 mg, 0.18 mmol) was added at 0 C, follo--wed by methylchloroformate (95.3 mg). 0.07 mmol). Stirring continued for another 0.5 h, then diluted with water and extracted with chloroform (100 mL 3•). After brine washing, the organic layer dried. sodium sulfate, concentra--ted. Purifying the residue on a silica gel column (60-120 mesh) with acetone-methanol (93:7 to 85:15) yielded a sticky substance, which was washed repeatedly. Diethyl ether and ethylacetate provide compound 24 as a colorless solid (70 mg, 23.4%). Mp 187–191 C. KBr: 3254, 1737, 1718, 1514, 890, 732 cm 1. 1H NMR (CDCl3, 200 MHz): 7.76 (d, (s, 1H, J = 8.5 Hz, 7.68), 7.59 (d, J = 8.8 Hz, 7.28 (br s, 1H), 6.31 (s, 2H), 6.28–6.25 (m, 2H), 4.79–4.72 (m, 1H), 4.39 (d, J = 2.7 Hz, 4H), 4.18 (t, J = 9.1 Hz, 1H), 3.84 (7.7 Hz, 1H), 3.41–3.30 (m, 4H). 13C NMR (50 MHz, CDCl3): d 157.2, 154.1, 136.6, 135.5, 129.7, 129.5, 121.5, 120.9, 120.1, 119.5, 109.4, 109.3, 71.4, 66.1, 59.6, 51.5, 47.2, 43.3. MS: 358 (M++1), 326. Anal. Calculated for C18H19N3O5: 60.48; 5.36; 11.76. Found: C, 59.95; H, 5.29; N, 11.90.

Etyl (S)-[3- (4H,6H-5-oxa-10b-aza-benzo [e] azulen- 8-yl) -2-oxo-oxazolidin-5- ylmethyl]-carbamate :

After replacing methyl chloroformate with ethyl chlorofor--mate, the title chemical was produced. Purified on a silica gel column (60-120 mesh) with acetone-chloroform (1:4 to 1:3) yielded a sticky chemical, washed with diethyl ether and ethylacetate to obtain a colorless powder in 29% yield. Mp 195 C. 3259 KBr IR, 3129, 2924, 1738, 1715, 1513, 1321 cm 1. 1.H NMR (CDCl3, 200 MHz): d 7.72 (d, J = 6.8 Hz, 1H), 7.60 (s, 7.45 (d, J = 8.8 Hz, 1H), 7.07 (s, 1H), 6.36 (s, 2H), 5.18 (br s, 1H), 4.83, 4.50 (d, 3.9 Hz, 4H), 4.10–4.17 (m, 3H), 3.92 (t, 7.8 Hz, 1H), 3.63 (d, J = 5.4 Hz, 2H), 1.22–1.29 (m, 3H). 13C NMR (50 MHz, DMSO-d6): d 156.7, 154.1, 136.7,135.5, 129.8, 129.5, 121.5, 120.9, 120.1, 119.5, 109.4, 109.3, 71.5, 66.1, 59.9, 59.6, 47.2, 43.2, 14.6. MS: 371 (M+), 342, 296, 154, 91. Anal. Calculated C19H21N3O5: C, 61.43; H, 5.70; N, 11.31; Found, 60.99; H, 5.33; N, 10.96.

Benzyl (S)-[3- (4H,6H-5- Oxa-10b-aza-benzo [e]azu-len-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-carba :

This chemical was produced by replacing methylchloro--formate with benzylchloroformate in the derivative 24 procedure. Purification on a silica gel column (60-120 mesh) with acetone-chloroform ‘1:4 to 3:7' yielded a cream-colored powder with 14% yield. Mp 198 C. IR (KBr): 3266, 3066, 2961, 2864, 1735, 1718, 1517, 1449, 1264 cm 1. 1H NMR (CDCl3, 200 MHz): d 7.68 (1H, J = 8.8 Hz). 7.55 (1H), 7.32–7.45 (6H), 7.05 (1H), 6.34 (1H). 2H), 5.25 (br s, 1H), 5.12 (s, 1H), 4.78–4.90 (m, 3H), 4.48 (d, J = 5.9 Hz, 3H), 4.10 (t, 9.0 Hz, 1H), 3.87 (t, 7.6 Hz, 1H), 3.59–3.65 (m, 2H). 13C NMR (50 MHz, DMSO-d6): d 156.7, 154.1, 136.9, 136.5, 135.5, 129.7, 129.5, 128.3, 127.8, 127.6, 121.5, 120.9, 120.1, 119.4, 109.4, 109.3, 71.5, 66.1, 65.5, 59.6, 47.1, 43.3, 40.7. MS: 433 (M+), 325, 296, 154, 108, 91. 109.3, 71.5, 66.1, 59.9, 59.6, 47.2, 43.2, 14.6. MS: 371 (M+), 342, 296, 154, 91. Anal. Calculated C19H21N3O5: C, 61.43; H, 5.70; N, 11.31; Found, 60.99; H, 5.33; N, 10.96.

S-N-[3-(4H,6H-5-Oxa-10b-aza-benzo[e]azulen-8-yl)-2-oxo-oxazolidin-5-ylmethyl]-thioacetamide :

Triethylamine (200 mg, 0.67 mmol) and ethyldithioacetate (43 lL) were added to amine 14 (200 mg, 0.67 mmol) in THF (2 mL) and agitated for 5–10 min. 6 h. The residue from volatile evaporation was purified on a silica gel column (60-120 mesh) using ethylacetate–petroleum ether (3:2) system. title chemical 27 (50 mg, 21%) white solid. Mp 196–198 C. KBr: 3251, 2919, 2851, 1749, 1619, 1589, 1511, 1321, 1066 cm 1. 1H NMR (CDCl3) 200 MHz): d 8.03 (1H br), 7.63 (2H t, J = 8.3 Hz). 7.43 (d, J = 8.3 Hz, 1H), 7.04 (s, 1H), 6.34 (s, 2H), 5.02 (d, 6.8 Hz, 1H), 4.48 (d, 4.4 Hz, 4H), 4.05–4.37 (m, 3H), 3.92 (t, 8.0 Hz, 1H), 2.62. 50 MHz 13C NMR (CDCl3 + DMSO-d6): d 201.3, 153.0, 135.3, 135.0, 129.0, 128.6, 120.2, 119.2, 118.4, 108.3, 108.2, 69.4, 65.5, 59.0, 46.9, 46.5, 31.9. MS: 357 (M+), 314, 313, 213, 154, 101.

O-Methyl S-[3- (4H,6H-5-oxa-10b-aza-benz [e] azulen-8-yl) -2-oxo-oxazolidin-5-ylmethyl]-thiocarbamate :

A solution of amine 14 (250 mg, 0.84 mmol) in THF was agitated for 6 hours at RT with the addition of carbon disulfide (127.3 mg, 1.67 mmol) and triethylamine (0.23 mL, 1.7 mmol). After adding methylchloroformate, the reaction mixture was agitated for 30–45 min. After diluting with 200 mL water, the aqueous layer was extracted with 300 mL 2• ethylacetate. The organic layer was brine-washed, dried, and concentrated. The residue was dissolved in methanol and recirculated for 16 hours. After evaporating volatiles, the chemical was purified using silica gel (60- 120 mesh) column with methanol–chloroform (1:9) eluent to yield compound 28 (135 mg, 40%). Colorless powder Mp 186–188 C. KBr IR: 3222, 3044, 2944, 1738, 1513, 1447, 1319, 1200, 1068, 723 cm 1. CDCl3, 200 MHz 1H NMR: d 7.70 (dd, J = 2.4 and 6.3 Hz, 1H), 7.58 (d), 7.43 (d, J = 8.3

Hz, 1H), 7.05 (s), 6.89 (br s), 6.34 4.95 (br s, 1H), 4.48 (d, J = 4.4 Hz, 4H), 3.90– 4.19 (m, 3H), 4.01 (s, 3H). 13C NMR (CDCl3, 50 MHz): d 192.8, 154.3, 136.7, 136.1, 130.5, 130.1, 121.6, 120.5, 120.4, 119.8, 109.7, 71.4, 66.9, 60.4, 58.6, 57.7, 47.5. MS: 373 (M+), 341, 312, 213, 117. Anal. Calculated for C19H21N3O4S: C 57.90, H 5.13, N 11.25. Found: C, 57.49; H, 5.07; N, 11.11.

O-Ethyl (S)-thiocarbamate :

The title chemical was produced in 38% yield utilizing ethanol instead of methanol as in compound 28. Colorless powder Mp 200 C. KBr: 3423, 2924, 1636, 1541, 1509, 1197, 714 cm 1. 1H NMR (CDCl3, 200 MHz): 7.68 (d, 7.59 (s, 1H), 7.44 (d, 8.3 Hz, 1H), 7.05 (1H), 6.70 (1H), 6.34 (2H), 4.93 (1H), 4.51–4.55 (m, 2H), 4.48 (d, J = 3.9 Hz, 4H), 3.96–4.19 (m, 4H), 1.26–1.36. 13C NMR (50 MHz, CDCl3): d 192, 154.2, 136.7, 136.1, 130.5, 130.1, 121.6, 120.5, 120.4, 119.8, 109.7, 71.4, 70.7, 68.2, 67.1, 66.9, 60.4, 47.5, 14.1. MS: 387 (M+), 343, 341, 213, 131, 102. Anal. C19H21N3O4S calcd: C 58.90; H 5.46; N 10.84. Found: C, 58.69; H, 5.71; N, 10.98.

Materials and procedures in biology :

1. Organisms. S. was one of the six reference Gram positive bacteria, both sensitive and resistant. Aureus ATCC 29213 (MSSA), S. aureus ATCC 49951 (Smith), S. MRSA, Entero--coccus faecalis ATCC 33591 29212, E. Enterococcus faecalis NCTC 12201 (VRE) and 12202 (VREf).

2. Susceptibility testing. According to NCCLS recommenda tions, 104–105 bacteria were transferred to Mueller Hinton agar (MHM-Difco) using a multi-point inoculating device to calculate MIC. Plates incubated at 35 C for 16–18 h. MIC refers to the lowest antibiotic concentration that prevents observable growth on agar plates.

3. Pharmacokinetics. A single-dose oral pharmacokinetics experiment was conducted on Swiss albino mice using selected compounds (MIC < 4 lg/mL) and Linezolid. DRF Institutional Animal Ethics Committee approved all animal experiments. Compound 3 & Linezolid were tested in Wistar rats for single-dose oral pharmacokinetics. In over night fasting studies, test chemical was given orally at 30 mg/kg in 0.25% carboxymethylcellulose. Blood was drawn from orbital plexus at 0.5, 1, 2, 4, 6- and 8-h plasma was bio-analyzed. The plasma concentration was measured by liquid liquid. extraction and HPLC–UV detection with 100 ng/mL quantification. Pharmacokinetic parameters were estimated from mean plasma concentration. analysis without compart--ments.

4. Infection systemic. Select compounds were tested in a mouse systemic infection model with Lin-ezolid as positive control. Growth of S overnight. aureus ATCC 29213 was inoculated onto fresh BHIB and incubated on a rotary shaker at 150 rpm for 2 h at 35 C. In Swiss albino mice weighing 19–21 g (n = 6), 0.5 mL of 100 • LD50 in 5% Hog Gastric Mucin (Difco) was administered intraperitoneally. Oral or subcutaneous administration of the test chemical suspended in 0.25% sterile CMC occurred 1 and 6 h postinfection. ED50 was obtained using linear regression.

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